Fiscal Year:	FY 2005	**Task Last Updated:**	FY 07/20/2005
PI Name:	Kashiwagi, Takashi Ph.D.		
Project Title:	Transition from Ignition to flame Growth under External Radiation in Three Dimensions		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Surface combustion, shs, fire safety and smoldering		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
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Space Biology Special Category:	None		
PI Email:	takashi.kashiwagi@nist.gov	**Fax:**	FY 301 975-4052
PI Organization Type:	GOVERNMENT	**Phone:**	301 975-6699
Organization Name:	National Institute of Standards and Technology		
PI Address 1:	Building and Fire Research Laboratories		
PI Address 2:	Mail Stop 8665		
PI Web Page:			
City:	Gaithersburg	**State:**	MD
Zip Code:	20899-8665	**Congressional District:**	8
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Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
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No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA GRC
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
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Key Personnel Changes/Previous PI:			
COI Name (Institution):	Mell, William E. (NIST)		
Grant/Contract No.:	NCC3-919		
Performance Goal No.:	2.4		
Performance Goal Text:	Develop methods, data bases, and validating tests for material flammability characterization, hazard reduction, and fire detection/suppression strategies for spacecraft and extraterrestrial habitats.		

A series of experiments and extensive numerical calculations was conducted to study localized spot ignition and the subsequent transition to flame spread over thin plastic and paper sheets in different gravity levels including zero gravity and normal gravity. The intended fire scenario is ignition and subsequent fire growth over a thin polymeric material close to a very limited heat source such as the momentary burning of a small overheated wire in various different gravities. The results are intended to help insure a fire safe environment in spacecraft, the International Space Station, and future lunar and Martian habitats. This project focuses on two separate regimes: one is the early stage of localized spot ignition and the other is a later stage of fire growth from that ignition. With a thin polymeric sheet, two ignition events are observed and predicted; one is ignition on the surface exposed to an external heat (frontside ignition) and the other is ignition over the backside surface (backside ignition). In microgravity, the frontside ignition delay time of a

Task Description:	polymeric material tends to monotonically increase with an increase in imposed flow velocity in the range simulating ventilation flows in these confinements. However, in normal gravity, in lunar gravity, or in Martian gravity, it is predicted that the frontside ignition delay time depends strongly on the sample orientation angle with respect to the gravitational direction due to buoyancy induced flow, even in a quiescent environment. Understanding the backside ignition process is important for fire safety because fire growth rate increases significantly with two simultaneous flames on both sides of the material. It is predicted that backside ignition occurs only after the formation of a small hole through the sample, after the sample is consumed locally. The formation of the hole has counteractive effects on both frontside and backside ignition; one is the consumption of the sample (tends to suppress both ignition) and the other is the fresh supply of air from the backside to the front side through the hole due to buoyancy induced flow (tends to favor ignition). Our preliminary calculation on fire growth shows that total heat release (which represents the size of a fire) from flame spreading over a thin polymeric sheet strongly depends on gravity level and the highest heat release would be between zero and normal gravity. Limited experimental results obtained in microgravity show that the size and the length of flame over a thin paper sheet drastically increases with an increase in external flow velocity. The objective is to conduct experiments on spot radiative ignition followed by transition to flame growth over different combustible solid surfaces (paper and plastics) in different gravity levels from microgravity to normal gravity. The transition phenomena of flame and sample surface temperature distribution are measured and the effects of sample thickness, oxygen concentration, imposed flow velocity, ignition configuration, and sample materials on the transition and flame growth characteristics are determined. In addition, numerical codes are developed for solving the time-dependent, three-dimensional Navier-Stokes equations to describe spot ignition and subsequent transition to flame growth with various variables such as imposed external flow velocity, sample material and thickness, and oxygen concentration with different pressures by keeping partial pressure of oxygen at 0.21 in different gravity levels from zero gravity to normal gravity. The codes are validated by comparing the predicted results with the experimental data to determine the accuracy of the predicted results. The validated numerical model could be used for specific fire scenarios in spacecraft, ISS, lunar and Martian habitats. The numerical code could be used to estimate time-dependent heat release rate from the onset of localized ignition followed by subsequent flame growth over plastic materials in various gravities and also in various oxygen concentrations at various atmospheric pressures. With the material thermal properties measured in normal gravity as inputs to the code, the calculated results could be used as guidelines for the selection of materials and also as a part of overall fire safety evaluation of spacecraft, ISS, lunar and Martian habitats.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The numerical code developed under this project is general enough so that it can apply to fire scenario in normal gravity including buoyancy flow induced by gravity. The results could be used for improving public fire safety.
Task Progress:	Numerical computations and a series of experiments were conducted in microgravity to study the ignition characteristics of a thin PMMA sheet (thicknesses of 0.2 mm and 0. 4 mm) using a CO2 laser as an external radiant source. Two separate ignition events were observed, including ignition over the irradiated surface (frontside ignition), and ignition, after some delay, over the backside surface (backside ignition). The results of our numerical study predict that frontside ignition delay time increases monotonically with an increase in imposed flow velocity in zero-gravity. However, in normal gravity, in lunar gravity, or in Martian gravity, it is predicted that the frontside ignition delay time depends strongly on the sample orientation angle with respect to the gravitational direction due to buoyancy induced flow even in a quiescent environment. Understanding the backside ignition process is important for fire safety because fire growth rate and heat release rate increase significantly with two simultaneous flames on both sides of the material. The backside ignition was achieved in two different modes. In the first mode, after the laser was turned off, the flame shrank and stabilized closer to the fuel surface. This allowed the flame to travel from the frontside to the backside through a small, open hole generated by the laser's vaporization of polymethylmethacrylate (PMMA). In the second mode, backside ignition was achieved during the laser irradiation. The numerical calculation simulating this second process predicts fresh oxygen supply flows from the backside gas phase to the frontside gas phase through the open hole, which mixes with accumulated hot MMA fuel vapor which is ignited as a second flame in the frontside gas phase above the hole. Then, the flame initiated from the second ignition travels through the hole to ignite the accumulated flammable mixture in the backside gas phase near the hole, attaining backside ignition. The first backside ignition mode was observed in 21 % oxygen and the second backside ignition mode for 35 %. The duration of the laser irradiation appears to have important effects on the onset of backside ignition. For example, in 21% oxygen, the backside ignition was attained after a 3 s laser duration but was not observed after a 6 s laser duration (within the available test time of 10 s in the drop tower). Longer laser duration might prevent two-sided ignition in low oxygen concentrations. The formation of a hole through the thin sample has two counteractive effects on the ignition process; one is reduction in fuel supply rate (suppresses ignition) and the other is an increase in oxygen supply from the back side to the irradiated side (promotes ignition). Heat release rate is the most important measure representing the size of fire. Therefore, we are starting to investigate the effects of different gravity on heat release rate of a simple material using our numerical code. Our preliminary results of flame shape and distribution of local heat release rate .At the slow imposed flow of 2 cm/s, only flame spreading upstream is predicted in the gravity level from zero to 0.001 g because the downstream flame does not survive due to the oxygen shadow effects of the upstream flame. With a further increase in gravity, it is predicted that two flames simultaneously travel on opposite directions; one upstream and the other downstream. The downstream flame is supplied by oxygen from further downstream due to weak buoyancy induced flow. At the gravity level of 0.01 g, buoyancy induced flow overcomes the imposed flow and the size of the downstream flame is about the same as that of the upstream flame. With a further increase in gravity level, both flames become smaller but more intense. Total heat release is calculated by integrating the local heat release rate distribution for each gravity level . The results show that the maximum total heat release is predicted at around 0.02 g based on the conditions used in the calculation. However, the maximum local heat release rate monotonically increases with an increase in gravity due to higher oxygen supply rate via stronger buoyancy induced flow at higher gravity. These results might strongly depend on the imposed flow velocity, sample orientation, and oxygen concentration. Furthermore, the fire scenario shown in Figure 3 is based on a time-dependent flame spreading phenomena. Therefore, time-dependent heat release might show different dependency on gravity. More numerical study on this topic is needed.
Bibliography Type:	Description: (Last Updated: 07/20/2005)

Abstracts for Journals and Proceedings	Nakamura, Y. Kashiwagi, T., Olson, S.L., Nishizawa, K., Fujita, O., Ito, K., "Two-sided ignition of a thin PMMA sheet in microgravity" Numerical computations and a series of experiments were conducted in microgravity to study the ignition characteristics of a thin PMMA sheet (thicknesses of 0.2 mm and 0. 4 mm) using a CO2 laser as an external radiant source. Two separate ignition events were observed, including ignition over the irradiated surface (frontside ignition), and ignition, after some delay, over the backside surface (backside ignition). The backside ignition was achieved in two different modes. In the first mode, after the laser was turned off, the flame shrank and stabilized closer to the fuel surface. This allowed the flame to travel from the frontside to the backside through the small, open hole generated by the laser's vaporization of polymethylmethacrylate (PMMA). In the second mode, backside ignition was achieved during the laser irradiation. The numerical calculation simulating this second process predicts fresh oxygen supply flows from the backside gas phase to the frontside gas phase through the open hole, which mixes with accumulated hot MMA fuel vapor which is ignited as a second flame in the frontside gas phase above the hole. Then, the flame initiated from the second ignition travels through the hole to ignite the accumulated flammable mixture in the backside gas phase near the hole, attaining backside ignition. The first backside ignition mode was observed in 21 % oxygen and the second backside ignition mode for 35 %. The duration of the laser irradiation appears to have important effects on the onset of backside ignition. For example, in 21% oxygen, the backside ignition was attained after a 3 s laser duration but was not observed after a 6 s laser duration (within the available test time of 10 s). Longer laser duration might prevent two-sided ignition in low oxygen concentrations. ignition , Jan-2005
Abstracts for Journals and Proceedings	Nakamura, Y., Kashiwagi, T. "Effects of sample orientation on nonpiloted ignition of thin poly(nmethyl methacrylate) sheet bu a laser – 1. Theoretical prediction" Non-piloted ignition processes of a thin poly(methylmethacrylate) (PMMA) sheet (0. 2 mm thick) by a laser beam as an external radiant source are investigated by the three-dimensional, time-dependent numerical calculation. The effects of the sample orientation angle on ignition delay time in quiescent air in a normal gravity environment and of imposed velocity in a microgravity environment are determined. The numerical model includes heat and mass transport processes with global one-step chemical reactions in both gas and solid phases. A simple absorption model based on Beer's law is introduced and bulk absorption coefficients are applied to the solid PMMA and evolved methymethacrylate (MMA). The PMMA sample surface is kept normal to the incident radiation at all sample orientation angles. In a zero gravity environment, ignition delay time increases with an increase in imposed flow velocity. In quiescent normal gravity, ignition delay time has a strong dependency on the sample orientation angle due to a complex interaction of the buoyancy induced flow containing evolved MMA with the incident laser beam. Without the absorption of the incident radiation by evolved MMA, ignition is not achieved. The most favorable ignition configuration is the ceiling configuration (face down horizontal sample irradiated by upward laser beam). The formation of a hole through the thin sample due to consumption has two counteractive effects on the ignition process; one is reduction in fuel supply rate and the other is an increase in air supply from the side opposite to the irradiated side by the buoyancy induced flow through the hole. Ignition , Mar-2005